We might reasonably call Everett's approach the "courageous" formulation of quantum mechanics. It embodies the philosophy that we should take seriously the simplest version of underlying reality that accounts for what we see, even if that reality differs wildly from our everyday experience. Do we have the courage to accept it?

This brief introduction to Many-Worlds leaves many questions unanswered. When exactly does the wave function split into many worlds? What separates the worlds from one another? How many worlds are there? Are the other worlds really "real"? How would we ever know, if we can't observe them? (Or can we?) How does this explain the probability that we'll end up in one world rather than another one?

All of these questions have good answers—or at least plausible ones—and much of the book to come will be devoted to answering them. But we should also admit that the whole picture might be wrong, and something very different is required.

Every version of quantum mechanics features two things: (1) a wave function, and (2) the Schrödinger equation, which governs how wave functions evolve in time. The entirety of the Everett formulation is simply the insistence that there is *nothing else*, that these ingredients suffice to provide a complete, empirically adequate account of the world. ("Empirically adequate" is a fancy way that philosophers like to say "it fits the data.") Any other approach to quantum mechanics consists of adding something to that bare-bones formalism, or somehow modifying what is there.

The most immediately startling implication of pure Everettian quantum mechanics is the existence of many worlds, so it makes sense to call it Many-Worlds. But the essence of the theory is that reality is described by a smoothly evolving wave function and nothing else. There are extra challenges associated with this philosophy, especially when it comes to matching the extraordinary simplicity of the formalism to the rich diversity of the world we observe. But there are corresponding advantages of clarity and insight. As we'll see when we ultimately turn to quantum field theory and quantum gravity, taking wave functions as primary in their own right,